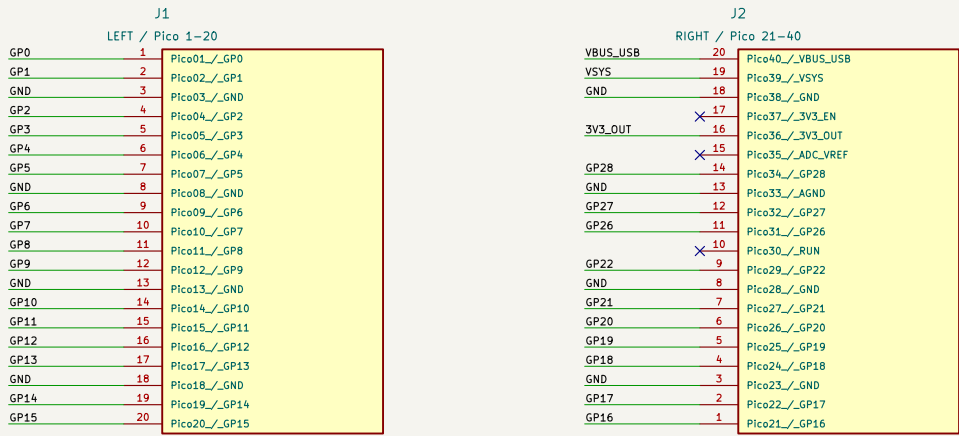


MAKE MUSIC – PROVISIONAL PICO CARRIER

Editable KiCad 10 schematic | Module order, parts, power and mechanical fit require primary integration

REMOVABLE PICO: TOP VIEW, USB AT TOP

Two 1x20 female sockets: 2.54 mm pitch, 17.78 mm row spacing. Carrier contacts are passive.



J1 pin 1 = Pico 1 (USB end). J2 pin 1 = Pico 21 (opposite USB).

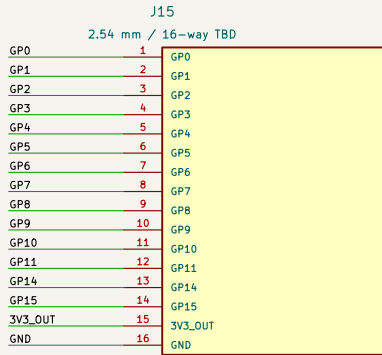
J2 pad 20 = Pico 40 at USB end: rotate J2 footprint 180 degrees relative to J1 in placement.

NC: Pico 30 RUN, 35 ADC_VREF, 37 3V3_EN. On-module defaults retained.

Pico 33 AGND tied to GND for digital touch use. GP23-25/29 are internal, not expansion.

SSQ candidate: 1.02 mm drills / 1.52 mm pads. Verify finished holes and male-header fit.

FREE GPIO EXPANSION – ORDER PROVISIONAL



GP0/GP1: optional I2C SDA/SCL.

No I2C pull-ups fitted until bus/device choice.

REVIEW ONLY – NOT FOR FABRICATION

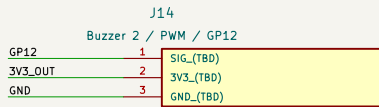
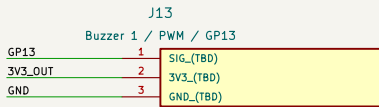
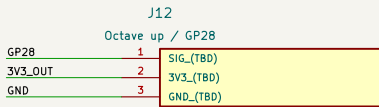
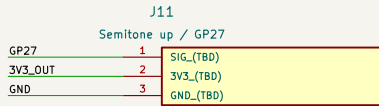
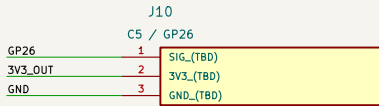
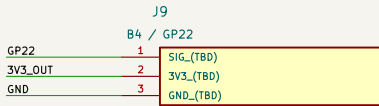
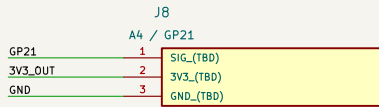
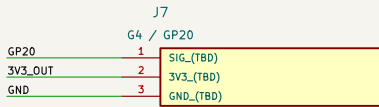
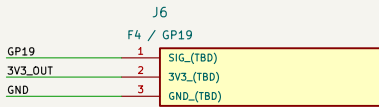
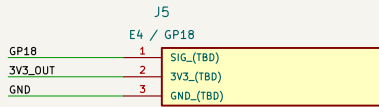
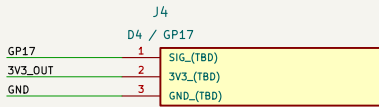
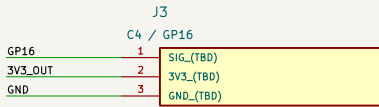
ERC checks carrier wiring only. Cell PCM, branch fault limits, module electronics and load/fit remain unverified.

Sources: Raspberry Pi Pico/Pico 2 datasheets, Figure 2 pin allocation; see ../reviews/schematic.md.

TEN TOUCH / TWO BUZZER MODULE INTERFACES

ALL J3-J14: assumed 1=signal, 2=3V3, 3=GND. Verify actual module pin order before assembly.

Footprints unassigned. Only 3.3 V-compatible modules with suitable output/driver circuits may be used.



SIGNAL / POWER LIMITS TO RESOLVE

Touch: TTP223 at 3.3 V, direct mode (TOG=0). Firmware may adapt to confirmed active-high/low.

Two ST0238 modules selected: low-level trigger; HIGH idle/startup and current budget need review.

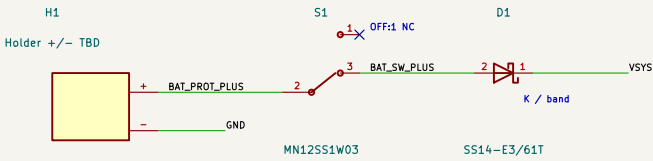
GP12 + GP13 share a PWM slice/frequency. Baseline firmware plays both at the same pitch.

Budget Pico 3V3_OUT current including 10 touch modules, 2 buzzer drivers and expansion loads.

J1/J2: SSQ-120-01-G-5 candidate assigned; mating fit, USB/BOOTSEL and battery clearances TBD.

Reference may be active-low (AHLB=1). Confirm actual polarity/pin order before firmware changes.

REV1 EXTERNAL-CHARGE BATTERY POWER



H1 +/-: logical identifiers, physical pads/polarity TBD.

Protected 18350 only: holder and cell fit not approved.

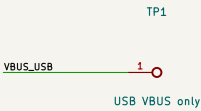
Branch fuse/current limit and PCM cutoff remain TBD.

S1 OFF Isolates battery; USB can still power Pico.

USB -> Pico onboard diode -> VSYS (inside Pico).

No carrier VBUS bridge; no onboard charger.

Remove cell for external charging; see power handoff.



Cell/L1 charger are review targets, not approved parts.